Reliable Taylor-Cone Angle Identification through Power-Law Exponent Crossing

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Abstract

Reliable Taylor-cone angle identification requires both an accurate field solution and an observable that distinguishes neighbouring candidate geometries. This study compares amplitude-projected stress-residual minimization with flank-field power-law exponent crossing in a reduced axisymmetric electrostatic–capillary model. Sparse finite differences with an immersed interface provide the numerical field solution; manufactured-solution and curved-boundary refinement tests assess parts of that workflow. The residual objective is shallow near its minimum, so distinct cone geometries can give similar values. By contrast, when the analytical Taylor potential is imposed on the far boundary, the candidate angle at which the fitted flank exponent crosses $p = -1$ lies between 49.07° and 49.25° across the tested settings. This is an ideal-limit consistency result, not an independent prediction of a physical cone angle: with a grounded outer boundary, the fitted exponent never reaches -1 and the crossing diagnostic is undefined. A synthetic-image round trip qualifies the extraction software on prescribed analytic silhouettes; a separate workflow applies image extraction and a fixed-pressure stress diagnosis to published cone-jet photographs. The latter is a secondary-data application, not validation of exponent crossing or a calibrated physical measurement. The result is a bounded comparison of angle observables and their boundary-condition dependence.

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1 Introduction

A Taylor cone is the conical liquid interface that can form when electric Maxwell stress competes with capillary stress. In the ideal charge-free, perfectly conducting limit, its classical half-angle is $\alpha_{\mathrm{T}} \approx 49.29^{\circ}$ [1]. The fluid dynamics of Taylor cones connect this static equilibrium to moving interfaces, charge transport, and cone-jet emission [2].

Interfacial stress-balance theory [3] and the Taylor–Melcher leaky-dielectric model [4] describe how electrical stresses interact with fluid motion and surface charge. Studies of emitted current [5] and cone-jet scaling [6] relate emission behavior to liquid properties and flow rate, while coupled cone-and-jet models calculate interface shape, electric fields, and surface charge [7]. Experiments on cone-jet operation [8] and classifications of spraying modes [9] further show the importance of specifying the operating regime when interpreting observed shapes.

Against this background, the present study examines the geometric information carried by two observables in a static model. Reliable angle identification requires a diagnostic that responds sufficiently to changes in candidate geometry. Comparing its sensitivity under controlled numerical and boundary settings provides a way to assess that requirement.

This paper addresses the following numerical question: once an electric field has been computed around a candidate cone, does the selected observable identify its half-angle? We compare the minimum of an amplitude-projected stress residual with the crossing of a fitted flank-field power-law exponent. The distinction matters because a well-resolved field does not by itself make an inverse geometric estimate unique: a shallow objective can assign nearly equal values to distinct geometries.

The solver is a two-dimensional axisymmetric finite-difference workflow with an immersed representation of the liquid interface (Figure 1). Manufactured solutions and curved-boundary refinement test components of the electrostatic calculation. Parameter and initialization studies then compare the two angle diagnostics. A separate image and stress-analysis workflow is demonstrated on published ethanol cone-jet figures, extending the analysis to secondary-data geometry extraction and fixed-pressure stress diagnosis.

Contributions.

1. We show that, for the tested restricted cone family, amplitude-projected stress residuals are weakly angle-discriminating: their minima move by several degrees with numerical choices and initialization while their values change little.
2. We show that, with the analytical Taylor potential imposed at the far boundary, the $p = -1$ flank-field crossing is substantially less variable across the tested configurations (49.07°–49.25°). This is a consistency check of the imposed Taylor far field, not an independent physical-angle prediction. With a grounded outer boundary, no $p = -1$ crossing occurs.
3. We document the numerical and image-processing evidence with its provenance: reproducible manufactured, refinement, multistart, and synthetic tests; a retained FEM comparison whose source and mesh are absent; and a published-image workflow whose extracted intermediates are absent.

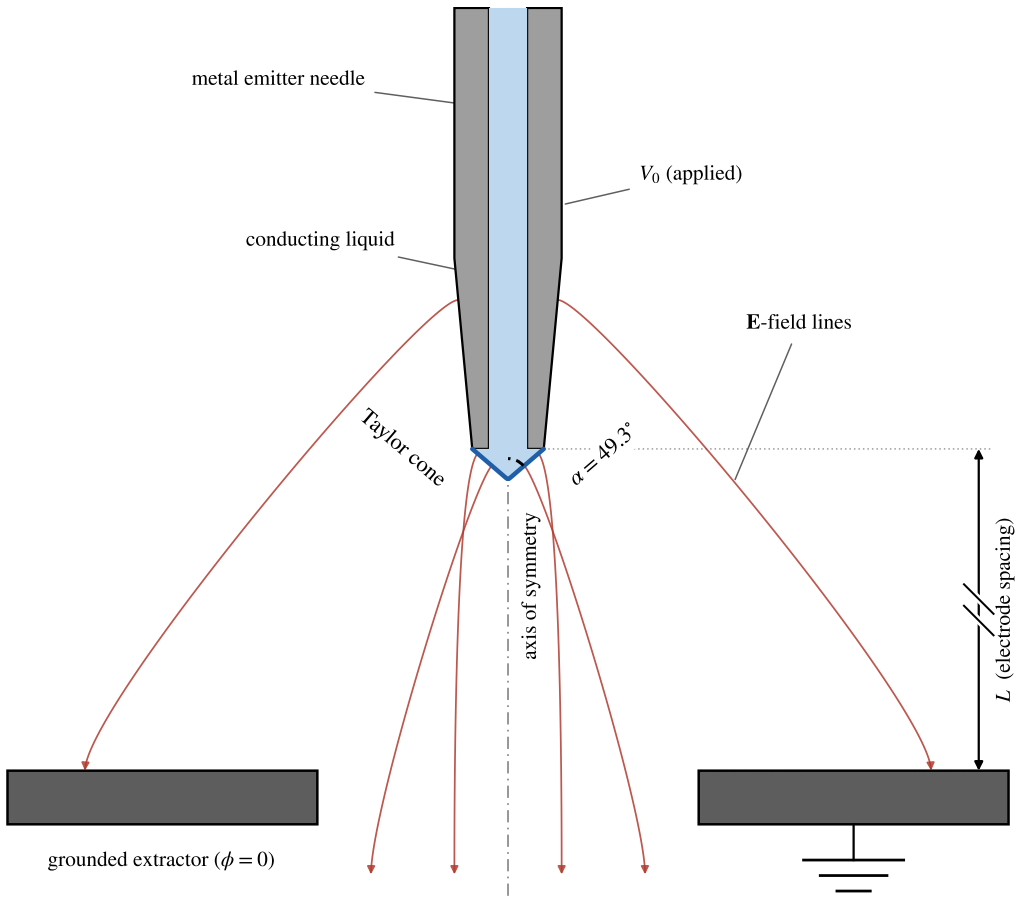


Figure 1: Axisymmetric electrostatic–capillary model (not to scale). A conducting liquid supplied at potential V_0 forms a candidate cone at the needle tip. The ideal Taylor half-angle is a charge-free, perfectly conducting limiting result; the finite-domain problems studied here impose either an analytical Taylor potential or a grounded condition on the outer boundary. The grounded extractor is at $\phi = 0$.

2 Research Question and Hypothesis

The central research question is how amplitude-projected stress-residual minimization and flank-field power-law exponent crossing differ as numerical diagnostics of a candidate Taylor-cone half-angle, and how boundary conditions and numerical settings affect each diagnostic.

The working hypothesis is that the residual objective is weakly angle-sensitive after pressure and amplitude are projected out, whereas a $p = -1$ crossing is more stable only in the idealized configuration with imposed Taylor far-field data. Numerical verification distinguishes discretization accuracy from the ability of either observable to identify an angle.

To examine this hypothesis, the study uses a static, axisymmetric electrostatic–capillary model and a restricted family of rounded cones. The numerical comparison focuses on angle identification within this family. The published-image section provides a complementary secondary-data workflow for extracting geometry and evaluating fixed-pressure stress balance.

3 Theoretical Model and Numerical Methodology

The model answers a focused question: once the electric field around a candidate cone is computed, which diagnostic identifies its half-angle more reliably? We compare a stress-balance residual with a field-scaling exponent. The model is static and axisymmetric, describing the electrostatic–capillary balance of prescribed candidate surfaces.

The conducting liquid is an equipotential surface represented by $r = R(z)$, parameterized by cone half-angle and rounded-tip radius. In the surrounding gas, the axisymmetric potential satisfies

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{\partial^2 \phi}{\partial z^2} = -\frac{\rho_e}{\varepsilon_0}, \qquad \mathbf{E} = -\nabla\phi. \tag{1}$$

The angle-diagnostic comparison is charge-free ($\rho_e = 0$). Nonzero manufactured sources are used separately to verify the numerical operator.

The potential is prescribed on the conductor, and symmetry requires $\partial_r \phi = 0$ on the axis. Two outer-boundary configurations are distinguished: an imposed analytical Taylor potential and a grounded outer boundary. Each defines a distinct electrostatic problem, and their effects on the angle diagnostic are tested explicitly. The rounded cap and a flank-sampling window keep the diagnostic away from the ideal apex singularity.

From the computed field, the stress balance is evaluated at the interface. For a perfectly conducting liquid, the gas-side normal field $E_n = \mathbf{E}\cdot\mathbf{n}$ produces an electric pressure that competes with capillary pressure:

$$p_E = \frac{\varepsilon_0}{2}E_n^2, \qquad p_\gamma = \gamma\kappa, \qquad \mathcal{R} = p_\gamma - p_E - \Delta p. \tag{2}$$

Here γ is surface tension, Δp is a spatially constant pressure offset, and κ is the sum of principal curvatures. For the graph interface,

$$\mathbf{n} = \frac{(1, -R_z)}{\sqrt{1+R_z^2}}, \qquad \kappa = \frac{1}{R\sqrt{1+R_z^2}} - \frac{R_{zz}}{(1+R_z^2)^{3/2}}. \tag{3}$$

The optimizer searches a restricted cone family parameterized by half-angle and rounded-tip radius. It fits the constant pressure offset and electric-stress amplitude for each candidate. Let $\langle\cdot\rangle_w$ denote an arclength-weighted mean over the sampled flank and let $e = E_n/V_0$ be the field per unit voltage for a nonzero reference voltage V_0. Define

$$a = p_\gamma - \langle p_\gamma\rangle_w, \qquad b = e^2 - \langle e^2\rangle_w. \tag{4}$$

The least-squares amplitude and projected objective are

$$u^* = \frac{\langle ab\rangle_w}{\langle b^2\rangle_w}, \qquad J = \sqrt{\langle (a - u^* b)^2\rangle_w}. \tag{5}$$

This requires $\langle b^2\rangle_w > 0$. A positive amplitude corresponds to a model balance voltage $\sqrt{2u^*/\varepsilon_0}$ for the prescribed candidate geometry. Pressure and amplitude fitting remove two modes of mismatch, allowing distinct angles to attain similar values of J. The sensitivity study evaluates this geometric ambiguity.

The second diagnostic uses the spatial scaling of the flank field. In the ideal Taylor limit, the potential near the apex has the separated form $\phi = A\rho^\nu P_\nu(\cos\vartheta)$, where ρ is distance from the virtual cone vertex. Consequently,

$$E_n^2 \propto \rho^{2\nu-2}, \qquad p_\gamma \propto \rho^{-1}. \tag{6}$$

Matching electric and capillary stress scaling requires $2\nu - 2 = -1$, hence $\nu = 1/2$ and $E_n^2 \propto \rho^{-1}$. The equipotential condition $P_{1/2}(\cos\vartheta_0) = 0$ gives the classical liquid half-angle $\alpha_T = 180^\circ - \vartheta_0 \approx 49.29^\circ$ [1]. This classical result supplies the reference scaling for the angle comparison.

For each candidate angle α, the numerical diagnostic fits the flank data using

$$\log E_n^2 = c + p(\alpha)\log\rho, \qquad p(\alpha_*) = -1. \tag{7}$$

The crossing angle α_* tests the spatial scaling of the field rather than the magnitude of a fitted stress residual. It is compared with the residual-minimizing angle under changes in grid, domain, cap radius, and fitting window.

Together, the two diagnostics distinguish field accuracy from angle identifiability. The residual measures the stress mismatch after pressure and amplitude fitting, while the exponent crossing evaluates field scaling under the specified boundary conditions. With imposed Taylor far-field data, the crossing provides an ideal-limit consistency diagnostic; for the tested grounded-boundary case, the fitted exponent remains away from -1. These properties define the comparison in Section 5.

4 Solver Architecture

The solver is a scientific workflow rather than a general multiphysics model. For each prescribed rounded-cone geometry, it constructs an axisymmetric grid, represents the conductor with either staircase or immersed Dirichlet treatment, solves the sparse Laplace or Poisson system, and reconstructs the gas-side field. Interface normals and curvature then give Maxwell and capillary pressures. The analysis reports either an amplitude-projected stress residual or a log–log flank fit of E_n^2 against distance from the virtual vertex (Figure 2).

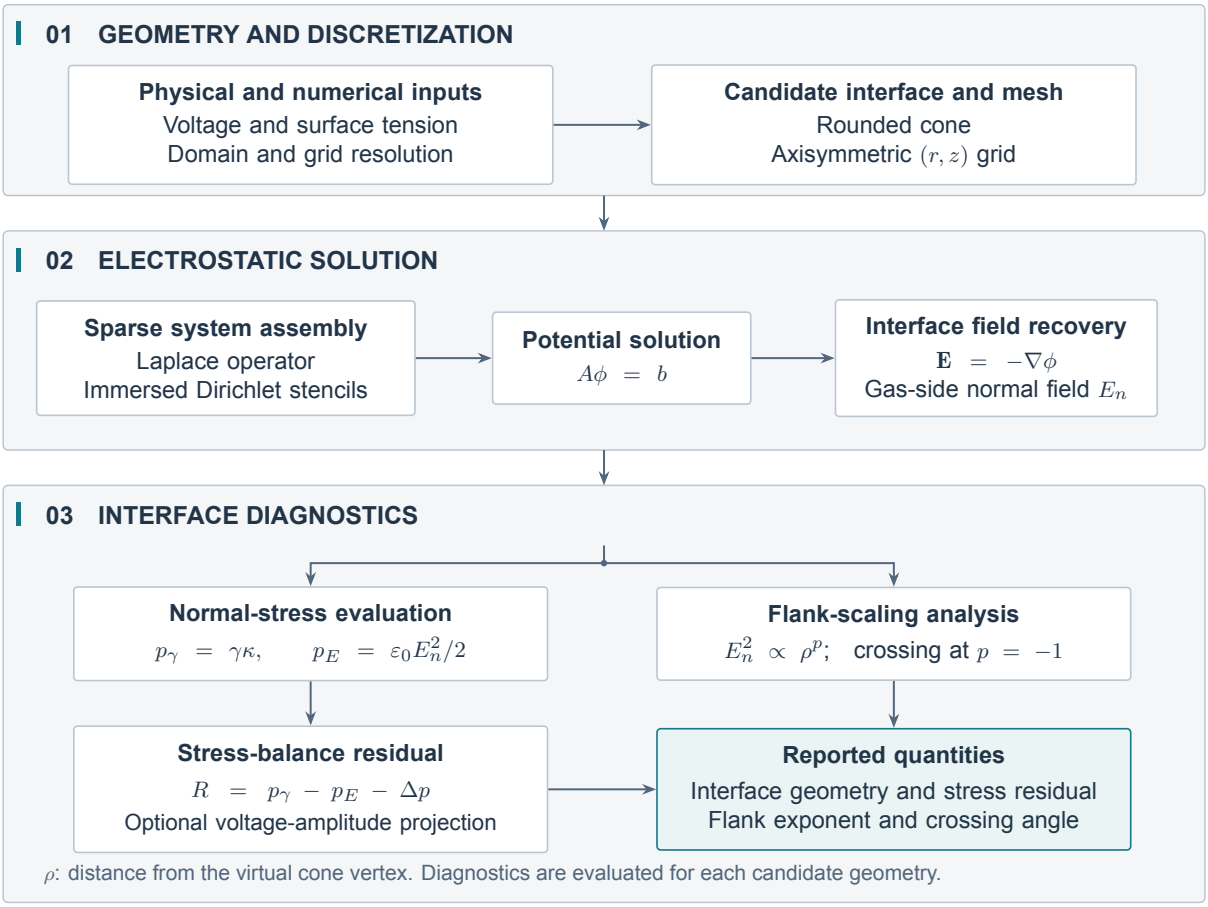


Figure 2: Scientific workflow for the charge-free immersed formulation. A prescribed rounded-cone geometry and boundary condition define the electrostatic solve; interface reconstruction then supplies either the projected stress residual or the flank-field exponent. The exponent crossing is evaluated only where the fitted exponent crosses -1.

The numerical core uses sparse matrices. Over the measured range of 675–85,073 points, the timing data for a single-threaded immersed Laplace solve follow an empirical $\mathcal{O}(N^{0.99})$ fit; a 121×177 grid took approximately 0.46 s on the machine recorded with the timing data. This fit describes the measured range, not an asymptotic complexity bound.

The associated image workflow renders analytic silhouettes for software tests and extracts contours, centreline registration, flank fits, and profile metrics. Its synthetic tests and its application to published photographs are kept separate: the former tests a known numerical round trip, while the latter provides a secondary-data geometry and stress analysis. Detailed module inventory is omitted from the main text, which is reserved for the numerical method and its verification.

5 Numerical Verification and Sensitivity

The tests below address distinct evidence questions: regular-grid operator accuracy, sensitivity to curved-boundary representation, angle-diagnostic behavior, and the image-extraction software path. The available results should not be read as interchangeable validation of the full physical model.

5.1 Regular manufactured Poisson problem

The source-derived quartic test uses $\phi = r^4 + z^4 + r^2 z^2$ and the axisymmetric source $18r^2 + 16z^2$. On grids 17×19, 33×37, 65×73, and 129×145, the recorded RMS errors were 1.82×10^{-3}, 4.45×10^{-4}, 1.10×10^{-4}, and 2.73×10^{-5}, with consecutive observed orders 2.035, 2.018, and 2.009. Error is the RMS difference between numerical and exact potentials at interior nodes; for halved mesh spacings, $q = \log(e_h/e_{h/2})/\log 2$. The finite-difference results support approximately second-order convergence for this regular manufactured problem. Figure 3 also retains a Q1 finite-element curve from an independent analysis. The FEM implementation, mesh, and figure-generation source are absent, so that comparison is not independently reproducible from the solver scripts; this evidence gap does not imply that the plotted values are known to be incorrect.

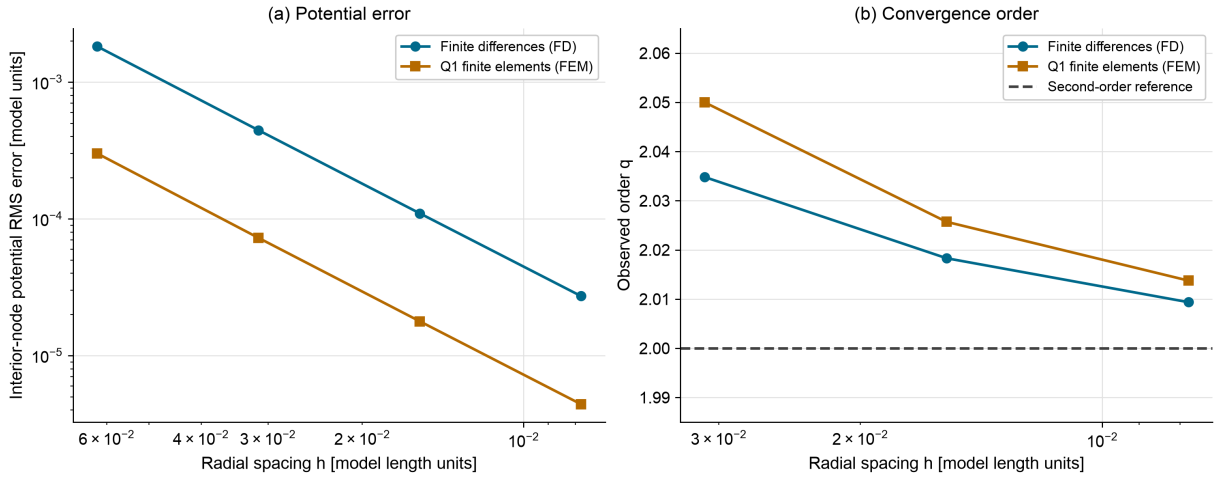


Figure 3: Regular-grid manufactured-solution convergence. Finite-difference errors are measured against the exact potential; the displayed Q1 FEM series is retained from an independent analysis whose source and mesh are not included. h is the geometric mean of radial and axial mesh spacings in model length units.

5.2 Curved-boundary refinement

The effect of curved-boundary representation is examined for the same rounded cone, with 30° half-angle and 0.5 mm apex radius, on grids of 31×45, 45×65, 61×89, and 87×129. Potential is evaluated at the same 56 gas-side locations on each grid. The inter-level change is

$$D_\ell = \left[\sum_{j=1}^{56} \left(\phi_\ell(\mathbf{x}_j) - \phi_{\ell-1}(\mathbf{x}_j) \right)^2 \right]^{1/2} .$$

For the immersed discretization, D_ℓ decreases from 10.334 to 4.264 and 2.549 V. The staircase values are 25.731, 7.107, and 11.267 V; the increase at the finest pair shows non-monotone inter-level variation. These values characterize refinement sensitivity for this setup, not a general convergence order for either curved-boundary method.

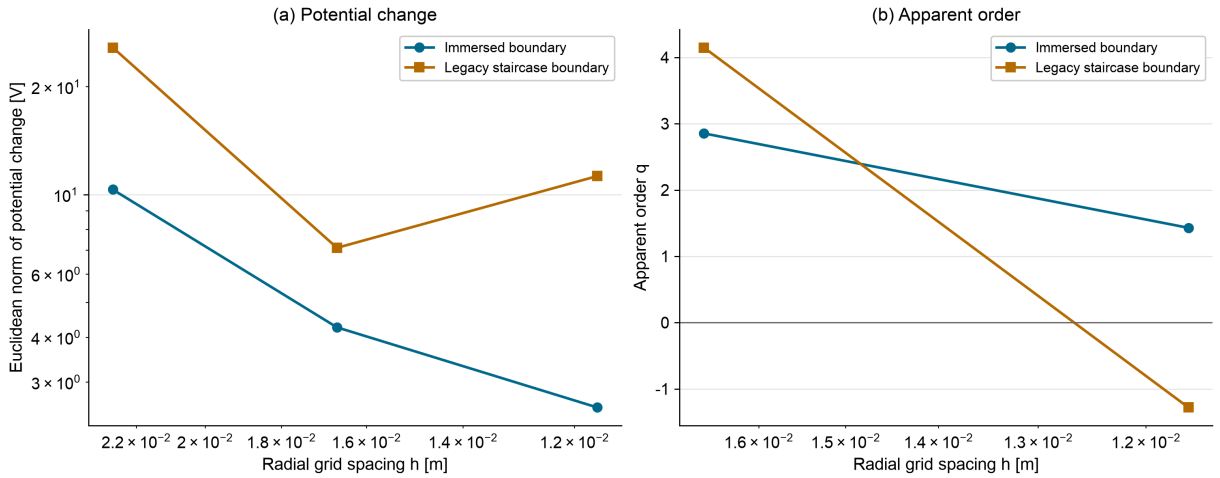


Figure 4: Curved-boundary refinement. Potential changes are evaluated at fixed gas-side points; the staircase sequence increases at the finest pair, so the displayed apparent orders are descriptive only.

5.3 Fixed-cone finite-element comparison

Figure 5 retains an immersed finite-difference (FD) versus body-fitted, piecewise-linear P1 finite-element (FEM) comparison for a charge-free problem. The plotted setup uses a domain radius and axial extent of 12.5 and 10 mm, a rounded cone of half-angle 49.29° with apex at $z = 8.6$ mm and cap radius 50 μm, a 1000 V conductor, and a grounded outer boundary. The FD grids are 65×97, 129×193, and 257×385; the fields are compared at common gas-side locations and at 41 interface positions over $z = 3$–6 mm.

The retained figure reports finest-level relative RMS differences of 0.024% for potential, 0.226% for vector field, 0.381% for interface-normal field, and 0.817% for Maxwell pressure. It also reports 1.071% for projected voltage and 2.518% for projected residual normalized by the RMS centered capillary pressure. The FEM code, mesh, and generating script are absent; accordingly these values are reported as an external analysis result, not as a comparison regenerated by the present project. Their absence limits traceability but is not evidence that the numerical values are wrong.

5.4 Angle and parameter sensitivity

Candidate half-angles from 46° to 52° are assessed across mesh resolution, domain size, cap size, flank-sampling window, and outer-boundary condition. Under the analytical Taylor far-field condition, the residual-minimizing angle lies roughly between 47.5° and 50°, while the fitted $p = -1$ crossing lies between 49.07° and 49.25°. These crossing values measure consistency with the imposed Taylor scaling; they are not independent predictions of a physical cone angle. With a grounded outer boundary, the fitted exponent shifts and does not reach -1, so the crossing diagnostic fails to return an angle. The plotted sensitivity figure is a retained campaign product: seven of its ten parameter series are reproducible with the current solver; the 25–55% and 35–65% flank-window series and the 1.50× domain series are not reproducible with the current exposed settings. This is a provenance limitation, not evidence that those plotted series are numerically incorrect.

Whereas Figure 6 evaluates one-dimensional angle sweeps at fixed cap radii, initialization and parameter-coupling sensitivity are tested separately via unconstrained two-parameter Powell optimization on the 31×45 grid using five starting pairs of half-angle and apex radius. With both

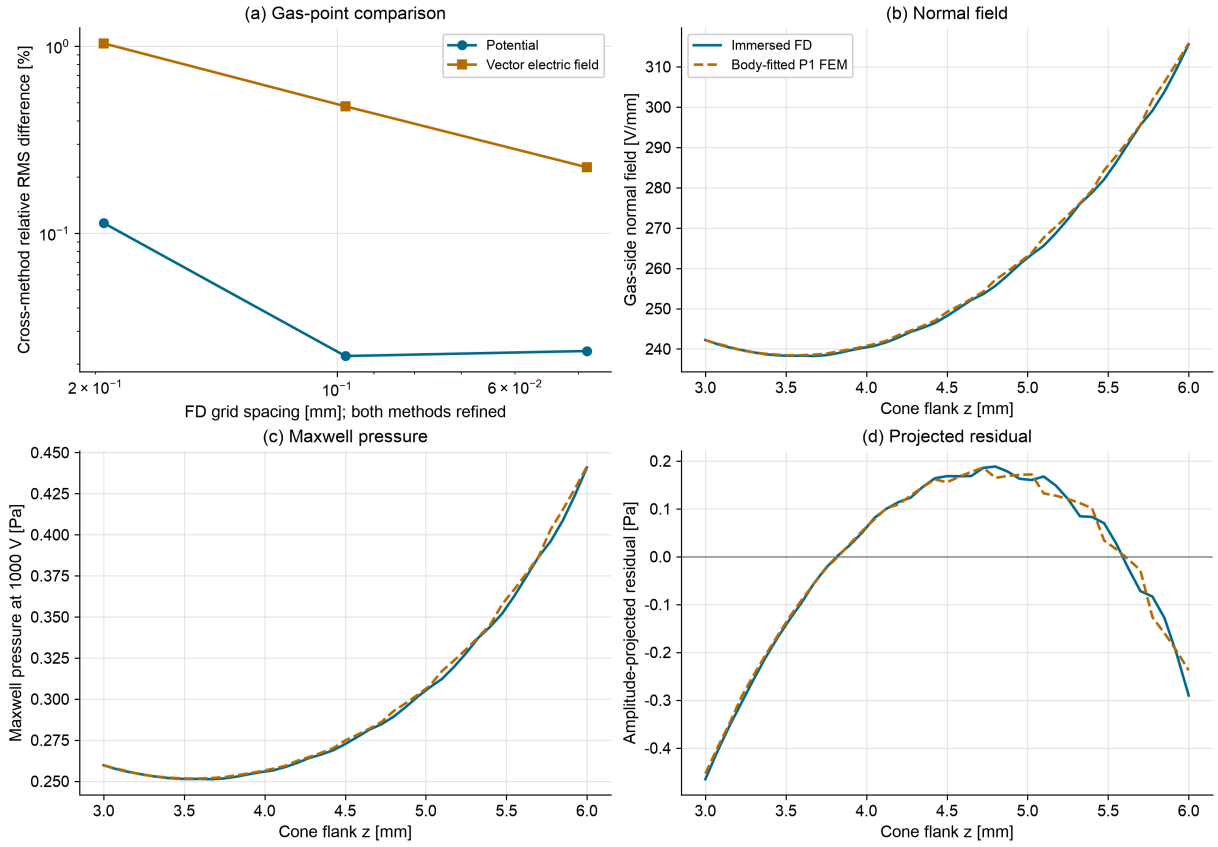


Figure 5: Retained fixed-cone FD–FEM comparison from an independent analysis. (a) Gas-point differences. (b) Interface-normal field. (c) Maxwell pressure at 1000 V. (d) Amplitude-projected residual. FEM source code and mesh are unavailable.

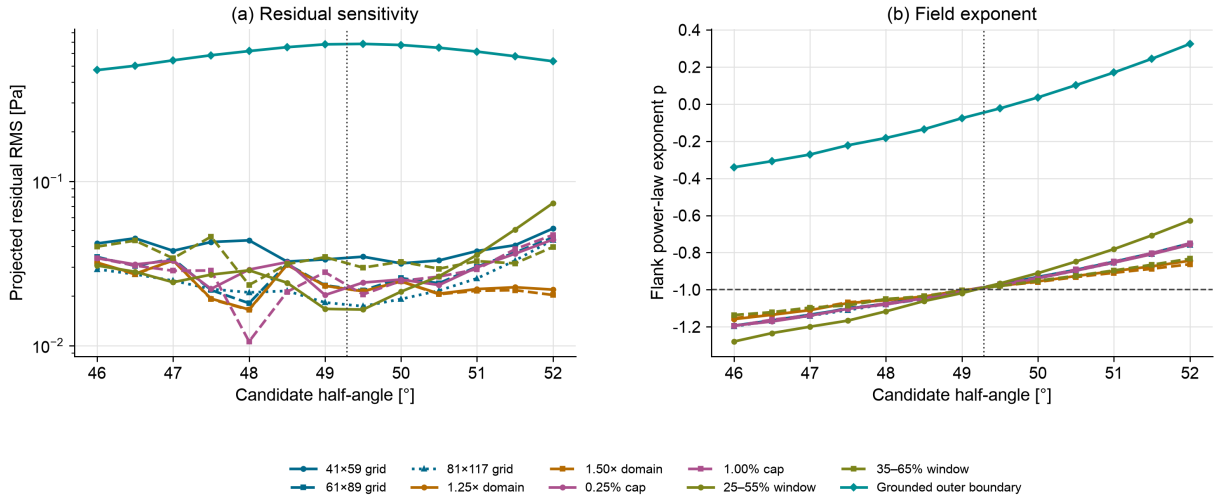


Figure 6: Angle sensitivity across retained campaign settings. The $p = -1$ crossing is available for the imposed Taylor far-field cases but absent for the grounded-boundary case. Three plotted parameter series cannot be regenerated from the current solver settings; see text.

parameters free, the converged half-angles fall to 40.20° to 42.79° and the apex-radius parameter ranges over 0.0704–0.0887 model length units (0.70–0.89 mm on the 10 mm model unit), while projected RMS residuals remain nearly flat between 0.01257 and 0.01333 Pa. This trade-off between tip curvature and flank angle demonstrates strong angle–radius ambiguity in the projected stress-residual objective.

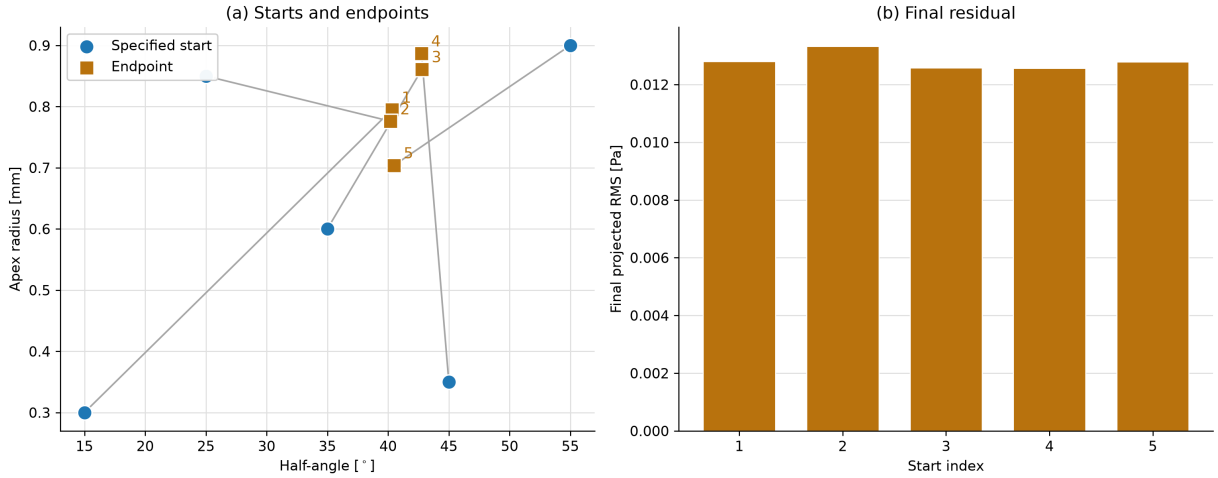


Figure 7: Multistart optimization. Starting and final parameter pairs are shown with their projected residuals; apex radii are plotted in millimetres, where one model length unit equals 10 mm.

5.5 Synthetic-image measurement test

The extraction pipeline is tested on analytic cone silhouettes with spherical-cap apices, rendered at 0.02 mm/pixel for a needle radius of 0.4 mm. The test crosses half-angles of 45°, 49.29°, and 55° with Gaussian noise standard deviations of 1.3, 4.0, 12.7, and 40.0 greyscale levels, labelled as nominal signal-to-noise ratios of 40, 30, 20, and 10 dB. Image scale and symmetry axis are prescribed by the renderer; the noise realization uses seed 20260922.

Over the two lowest noise levels, the largest angle error is 0.027° and the largest profile nRMSE is 0.62%; all cases at these levels meet the preset 1° angle and 2% profile limits. Across all twelve cases, angle error remains below the angle limit, but at the highest noise level the 55° silhouette has 10.5% profile nRMSE. The nominal SNR labels are rounded labels for prescribed noise levels, not exact SNR values.

These results qualify the numerical image-processing round trip, not measurement uncertainty on recorded images: the fitted profile and reference profile share the same analytic source, and the test does not include illumination variation, contact-line pinning, optical distortion, or recorded-frame resolution. Section 6 applies the workflow separately to published photographs. Its source photographs are cited, but the reduced images, extracted contours, and analysis intermediates needed to regenerate its reported measurements are not included.

6 Published-Photo Image and Stress Workflow

This section is a separate secondary-data application, not a validation of the exponent-crossing diagnostic or a calibrated physical measurement. It analyzes previously published ethanol cone-jet photographs and reported angles from Yang et al. [10]; no new laboratory data are presented. The

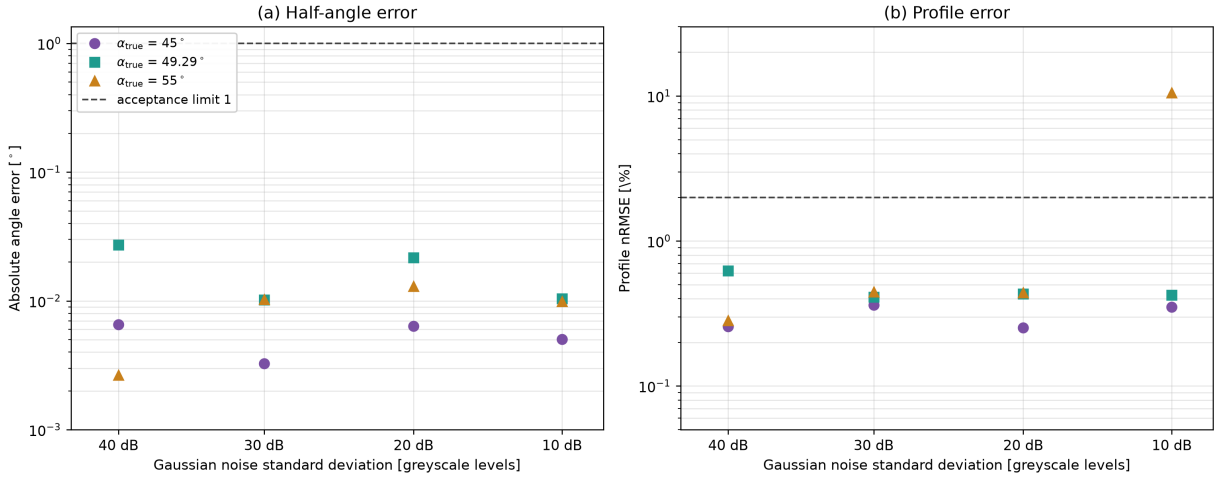


Figure 8: Synthetic-image extraction test. Markers show analytic-image cases; dashed lines mark the preset 1° angle and 2% profile limits. Noise-level SNR labels are nominal. Both axes are logarithmic.

cited source figures are available through that publication, but the reduced source images, extracted contours, metadata, and scripts that generated the following overlays and values are not included. The results are therefore retained for interpretation, not claimed to be reproducible from the solver scripts.

6.1 Image geometry and extraction

The source study reports a flow rate of $Q = 0.10$ μL/s, nozzle-to-plate spacing $H = 2$ mm, and aperture diameter $D = 2$ mm. The photographs are taken from its Figure 5; full cone angles are digitized from its Figure 6 and converted to half-angles using $\alpha = \theta/2$. The numerical representation uses the reported geometry together with prescribed outer-domain, exposed-needle, and tip-regularization settings.

Table 1: Published inputs and prescribed numerical settings for the separate image and stress workflow.

Quantity	Value	Source/status
Nozzle outer / inner diameter	0.30 / 0.16 mm	Yang et al., Sec. 1
Plate outer / aperture diameter	80 / 2 mm	Yang et al., Sec. 1; Fig. 6
Nozzle-to-plate spacing	2 mm	Yang et al., Fig. 6
Ethanol flow rate	0.10 μL/s	Yang et al., Figs. 5–6
Temperature; surface tension	25 °C; 0.022 N/m	Yang et al., Table 1
Image scale bar	300 μm / 191 pixels	Yang et al., Fig. 5
Outer-domain radius	80 mm	Prescribed numerical setting
Exposed needle length	5 mm	Prescribed numerical setting
Tip radius; local cell size	6 μm; 4 μm	Prescribed numerical setting
Pressure calibration voltage	1.7203 kV	Lowest reported point

An axisymmetric (r, z) representation places the nozzle exit at $z = 0$ and the grounded annular plate at $z = 2$ mm. The aperture and plate outer radii are 1 and 40 mm, respectively (Figure 9). The liquid surface and nozzle are prescribed at the applied voltage.

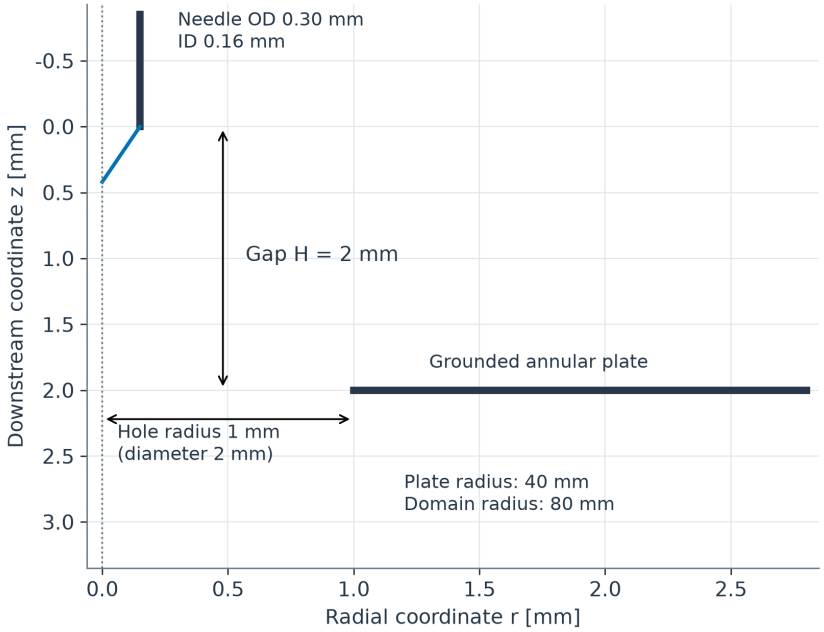


Figure 9: Axisymmetric representation for the published-data workflow. Plate and aperture dimensions are reported inputs; the outer-domain radius and tip treatment are prescribed numerical settings.

Four photographs at 1.80, 1.95, 2.10, and 2.30 kV are analyzed with the source scale bar, $300/191 = 1.57068$ μm/pixel. A nominal grayscale threshold of 110, paired left and right boundaries for centreline registration, and linear flank fits over 30–75% of visible height yield the displayed half-angles. The virtual tip aligns profiles; visible height is measured from nozzle exit to observed tip.

The retained extraction increases from 22.26° to 41.64° as voltage increases from 1.80 to 2.30 kV, while visible height decreases from 301.6 to 147.6 μm (Table 2). The registered profiles show this widening and shortening.

6.2 Comparison with reported angles

For the three pairs matched within 0.005 kV (1.80, 1.95, and 2.10 kV), the retained absolute half-angle differences are 1.970°, 1.520°, and 0.520°, yielding a mean absolute difference of 1.337° and RMS difference of 1.468°. These numbers compare two image-derived series from the published study; they are not a calibrated error estimate for this pipeline.

6.3 Extraction sensitivity and fixed-pressure diagnosis

Repeating the retained extraction at thresholds of 90, 110, and 130 gives a maximum half-angle range of 0.204° for the matched cases and 0.322° at 2.30 kV. Nominal straight-flank fit distances are 1.082, 0.837, 0.848, and 0.579 μm in ascending voltage order. Threshold sensitivity, fit distance, and discrepancy between published series are different quantities.

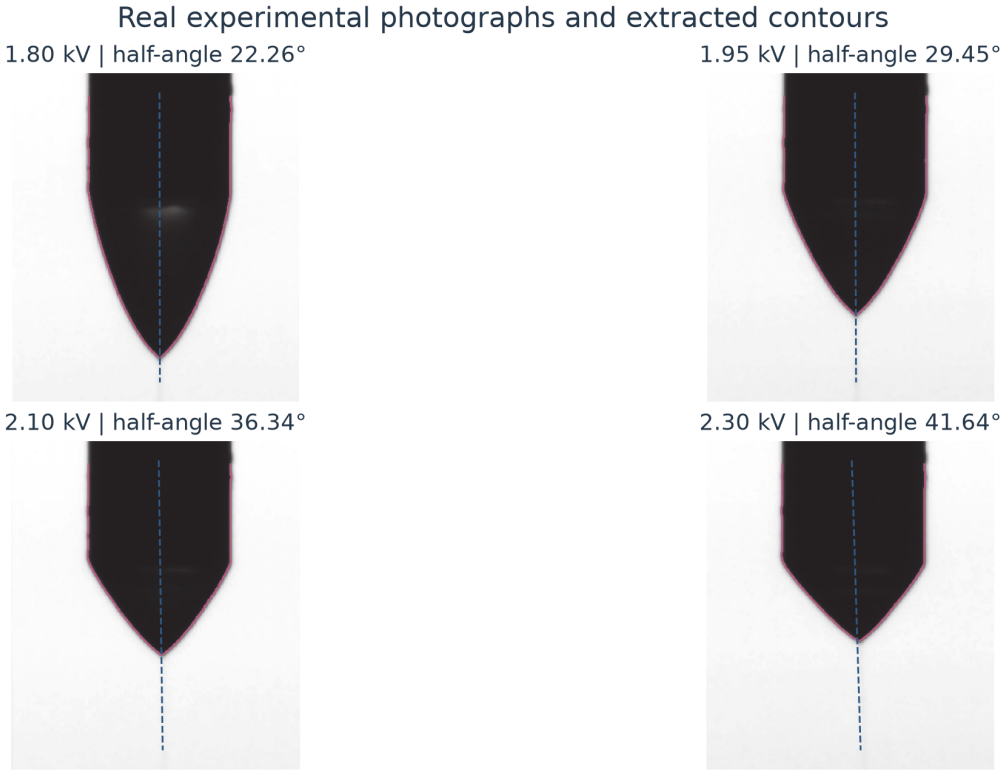


Figure 10: Published ethanol photographs with retained contour and axis overlays. Photograph source: Yang et al., Figure 5. The underlying image-processing intermediates are unavailable.

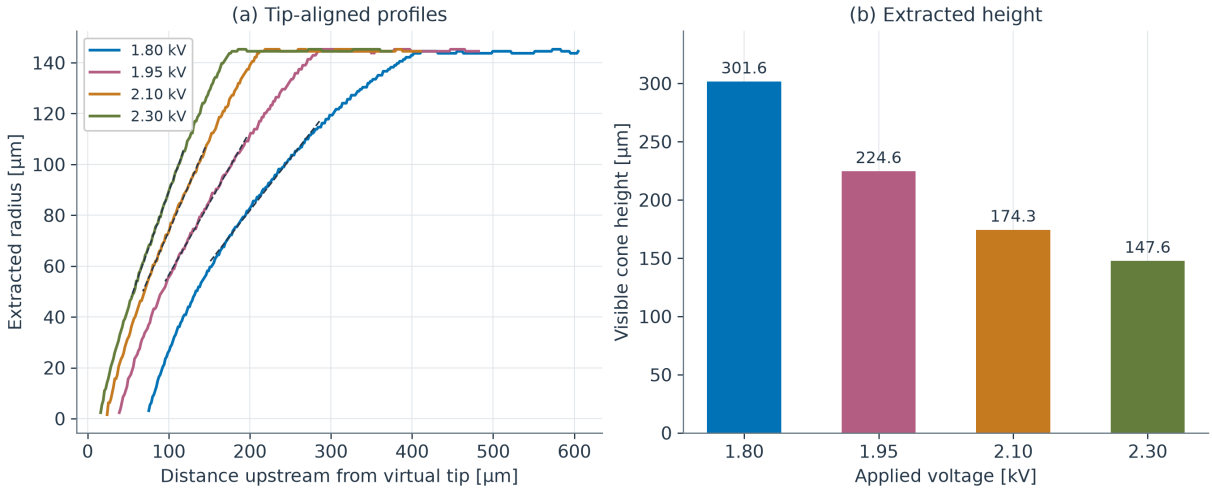


Figure 11: Retained registered profiles and visible heights from the published-photo workflow. Dashed lines summarize flank fits.

Table 2: Retained photo-derived and reported half-angles.

Photo V (kV)	Reported V (kV)	Photo α (°)	Reported α (°)	$\lvert\Delta\alpha\rvert$ (°)	Photo height (μm)
—	1.7203	—	19.80	—	—
1.80	1.8005	22.26	24.23	1.970	301.6
1.95	1.9503	29.45	30.97	1.520	224.6
2.10	2.1000	36.34	36.86	0.520	174.3
—	2.2503	—	42.99	—	—
2.30	—	41.64	—	—	147.6

Dashes indicate an unpaired source observation. The reported series and photographs come from separate figures in the same study.

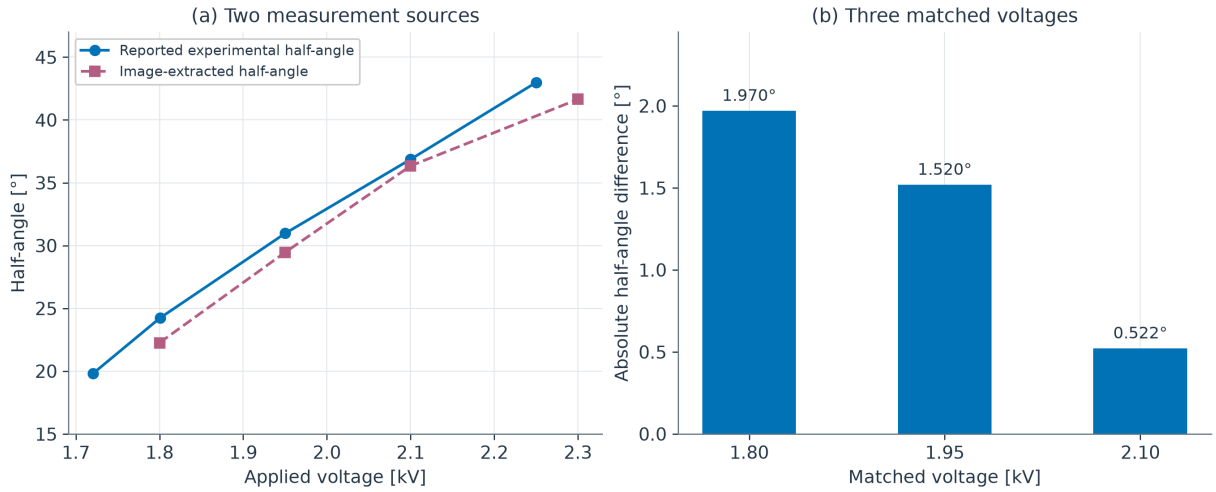


Figure 12: Published-series comparison. Photo extraction and reported markers are distinguished; differences use the three matched voltages. This plot is a consistency comparison, not solver validation.

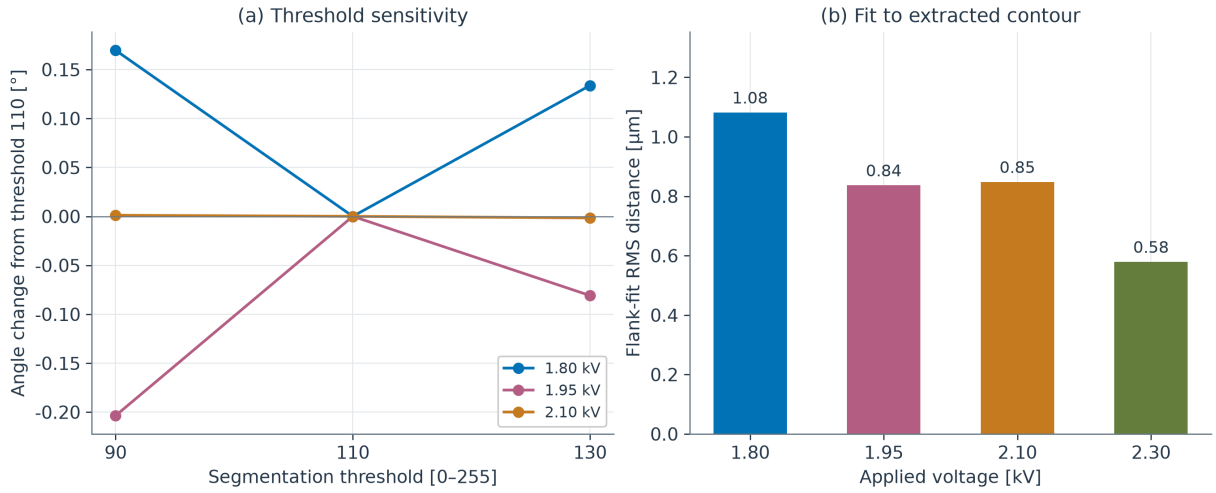


Figure 13: Retained extraction sensitivity: threshold-induced angle changes and nominal straight-flank fit distances.

For a fixed-pressure stress diagnosis, the reported half-angle at each voltage is used as a geometric input. Over the central 25–75% of cone height,

$$R(z;V) = \gamma\kappa(z) - \frac{\varepsilon_0}{2}E_n(z;V)^2 - \Delta p, \qquad \overline{R} = \langle R \rangle_A.$$

The pressure offset is calibrated at 1.7203 kV,

$$\Delta p = \left\langle \gamma\kappa - \frac{\varepsilon_0}{2}E_n^2 \right\rangle_{A,V=1.7203\,\mathrm{kV}} \simeq 6.458 \text{ Pa},$$

and then held fixed. The retained mean residuals are approximately 0, −44, −123, −201, and −278 Pa across the five reported conditions. They show that this fixed-pressure static closure does not balance the selected observed-angle series at the higher voltages. The displayed variations in local spacing, exposed-needle length, and tip radius preserve the negative trend; they do not establish the cause of the mismatch or validate a dynamic physical model.

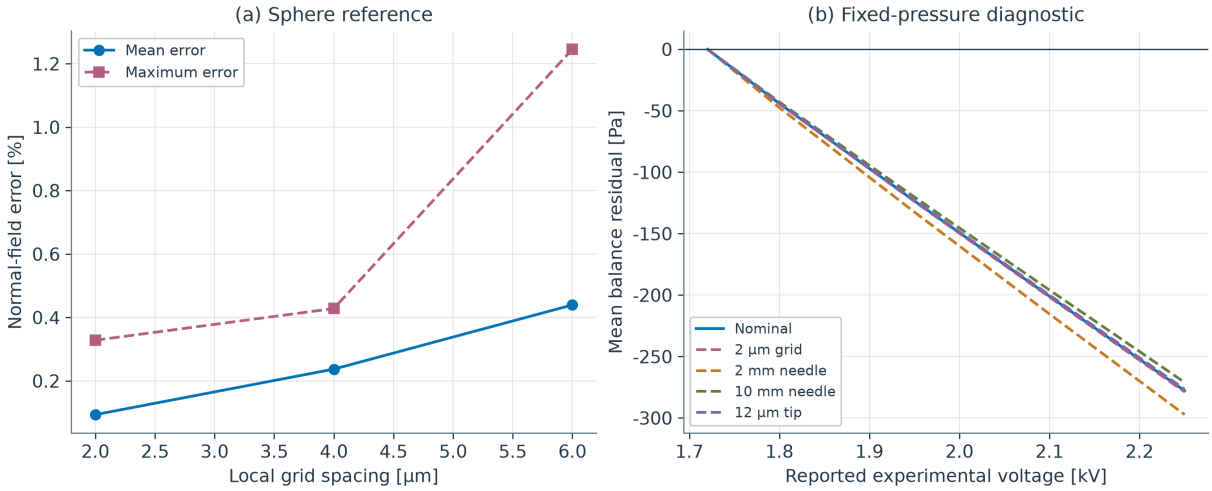


Figure 14: Retained field and fixed-pressure balance checks. (a) Sphere-reference normal-field errors. (b) Mean residuals at reported angles for five numerical settings. The analysis tests the stated closure only.

7 Limitations

The model is static, axisymmetric, and reduced-order. It does not solve transient fluid motion, leaky-dielectric surface-charge conservation, emission-current laws, breakup, plume dynamics, or thruster performance. The solver additionally implements phenomenological space-charge closures for studying Poisson shielding trends; these are not predictive emission models and are not used in the charge-free angle comparison.

The exponent crossing has a deliberately narrow interpretation. It is stable only in the tested configurations that impose the analytical Taylor far-field condition, and thus checks consistency with that imposed ideal limit rather than predicting a physical cone angle independently. In the grounded-boundary configuration, the fitted exponent does not reach −1 and no crossing angle exists. The residual diagnostic remains available, but its shallow minimum and initialization sensitivity prevent it from uniquely identifying a geometry in the tested restricted family.

Evidence availability limits traceability. The regular manufactured test, curved-boundary refinement, multistart study, and synthetic-image test can be reproduced from the solver scripts. The FEM source and mesh for the retained fixed-cone comparison are absent. The published-photo workflow cites its source study, but its reduced images, contours, and analysis intermediates are absent. These absences prevent regeneration; they do not show that the retained values are wrong.

Synthetic silhouettes assess a prescribed software round trip, not uncertainty on recorded images. The published-photo analysis is therefore a secondary image and stress application, not physical validation. No new physical testing is included or planned in this submission.

8 Conclusion and Future Work

Within the tested reduced model, the amplitude-projected stress residual does not uniquely identify a cone angle: distinct angle–radius combinations and numerical settings can give similarly small residuals. With the analytical Taylor potential imposed at the far boundary, the fitted flank exponent crosses $p = -1$ at 49.07°–49.25°, close to the ideal 49.29° Taylor half-angle. This is a consistency result for the imposed far-field condition, not an independent physical-angle prediction. Its scope is made explicit by the grounded-boundary case, where the exponent never reaches -1 and the crossing diagnostic returns no angle.

The reproducible numerical evidence supports approximately second-order behavior for the regular manufactured Poisson problem, characterizes curved-boundary refinement sensitivity, documents residual ambiguity under multistart optimization, and qualifies an analytic-silhouette extraction round trip. The retained FEM comparison and published-photo workflow add context but cannot be regenerated because their required artifacts are absent. The latter is a separate secondary-data image and fixed-pressure stress analysis, not validation of exponent crossing or full physical behavior.

Priority future work is to release the FEM mesh and source, archive reduced published-image inputs and extracted contours with their processing script, and regenerate every sensitivity series from declared settings. The next computational step is to test independently specified boundary conditions and compare the static model with higher-fidelity numerical solutions. Further secondary-data analysis would require archived published-image inputs and documented uncertainty; no new physical testing is proposed.

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